

AP Physics 2: Algebra-Based The Complete Exam Guide

All 7 units explained, from thermodynamics to modern physics, the four free-response question types, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.



3 hrs

85 min multiple choice
+ 95 min free response

42+4

multiple-choice +
free-response questions

7 units

thermodynamics to
modern physics

72–73%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, one score

Your AP score (1–5) combines a multiple-choice section and a free-response section, each worth half. A calculator and an equation sheet are allowed on the whole exam.

50%

Section I • Multiple choice • 42 questions

50%

Section II • Free response • 4 questions

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	50% of score 1 hr 25 min 42 questions	Single questions and short sets built around a stimulus such as a diagram, graph or data table. Calculator allowed.
II • Free response	50% of score 1 hr 35 min 4 questions	Four question types: Mathematical Routines • Translation Between Representations • Experimental Design and Analysis • Qualitative/Quantitative Translation.

HYBRID DIGITAL

Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

PROVIDED ON EXAM DAY

Equation sheet + calculator

A table of equations and constants is provided, and a calculator is allowed throughout. Knowing **when** an equation applies, and its conditions, matters more than memorizing it.

How to use this guide

Just starting?

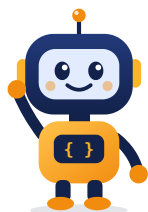
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 22 minutes, then use the final-four-weeks plan.



Pigo's tip: Physics answers are scored on **reasoning**. Start every explanation from a principle, such as Newton's second law or conservation of energy, then apply it to the situation.

SECTION I

02 The multiple-choice section

42 questions in 85 minutes gives you about 2 minutes per question. Many questions test reasoning with symbols and graphs rather than plugging in numbers.

42

questions, some in stimulus-based sets

~2:00

minutes per question on average

50%

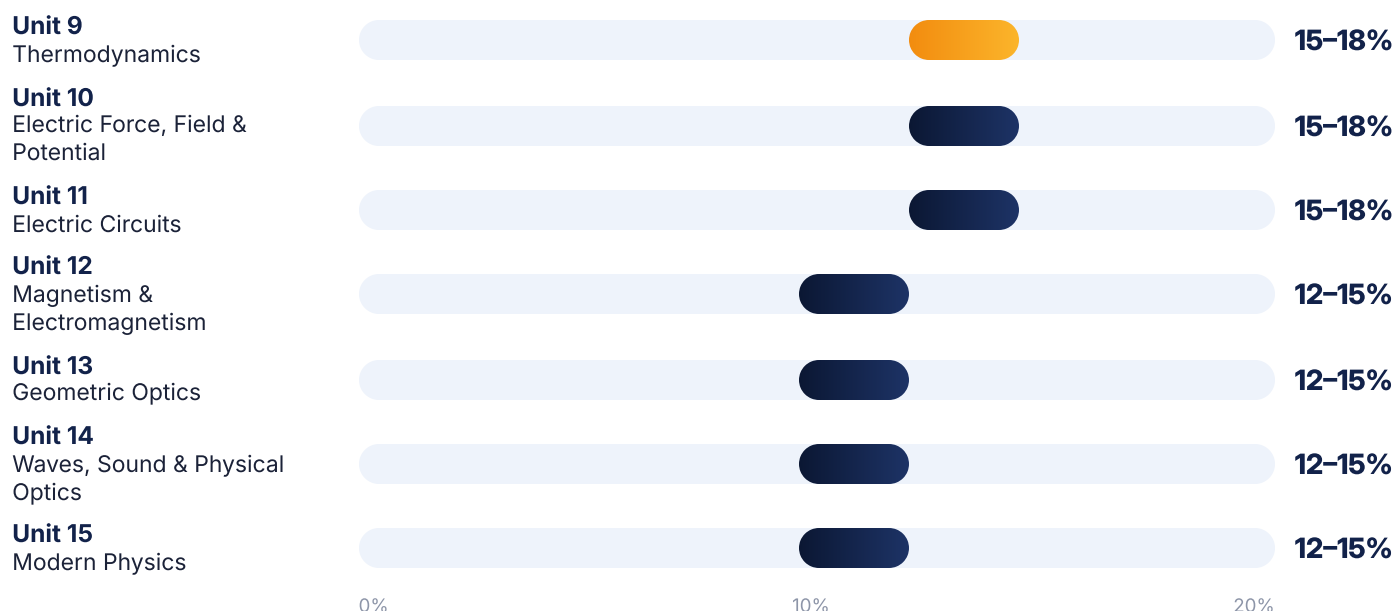
of your total AP score

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the multiple-choice section. Bars show minimum to maximum.



What this means for your study time: The seven units are close to equal, so there is no unit you can skip. Electricity (Units 10–11) together is nearly a third of the section.

Strategies that work on this section

- **Draw a diagram** for every question: forces, fields, circuits or rays. It takes seconds and prevents sign errors.
- **Check limiting cases.** What happens if a mass is zero or an angle is 90° ? Wrong choices often fail this test.
- **Use units to eliminate** answer choices with the wrong dimensions.
- **Read graphs as relationships:** slope and area under the curve both have physical meaning.
- **Work symbolically first**, then plug in numbers only if needed.
- **Flag and move on** after 3 minutes. Return once the quick points are banked.

03 WHAT THE COURSE COVERS

The 7 units, in plain English

Everything on the exam fits into these units.

UNIT

9

15–18%

Thermodynamics

Be ready to: use the ideal gas law and kinetic theory; apply the first law ($\Delta U = Q + W$) to PV diagrams; analyze heat engines and efficiency; describe thermal conduction and entropy.

UNIT

10

15–18%

Electric Force, Field & Potential

Be ready to: apply Coulomb's law; draw field lines and equipotentials; calculate field and potential from point charges; relate field to potential difference; analyze capacitors.

UNIT

11

15–18%

Electric Circuits

Be ready to: use Ohm's law and resistivity; reduce series and parallel networks; apply Kirchhoff's rules; calculate power; analyze RC circuits qualitatively.

UNIT

12

12–15%

Magnetism & Electromagnetism

Be ready to: find forces on moving charges and currents; use right-hand rules; describe fields of wires; apply Faraday's and Lenz's laws.

UNIT

13

12–15%

Geometric Optics

Be ready to: apply reflection and Snell's law; find critical angles; draw ray diagrams; use the thin lens and mirror equations and magnification.

UNIT

14

12–15%

Waves, Sound & Physical Optics

Be ready to: describe wave properties and standing waves; apply the Doppler effect; analyze double-slit and thin-film interference and diffraction.

UNIT

15

12–15%

Modern Physics

Be ready to: use photon energy and the photoelectric effect; read energy-level diagrams; describe wave-particle duality; balance nuclear decays and use $E = mc^2$.

Units are numbered 9–15 because they continue on from AP Physics 1 (Units 1–8). Physics 1 ideas such as energy and forces are used throughout.

04 SECTION II The 4 free-response questions

95 minutes, 4 questions. The four question types are the same every year, so you can practice each on purpose.

MR	TBR	LAB	QQT
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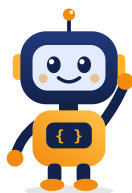
Mathematical Routines and Lab questions are typically worth the most points

MR Mathematical Routines Derive and calculate Use equations and principles to derive expressions and calculate quantities, often starting symbolically. Start from a fundamental equation and show each substitution.	TBR Translation Between Representations Diagrams, graphs, equations Draw a free-body diagram, energy bar chart or graph, then write equations that are consistent with it. Arrows on free-body diagrams start on the object and are labeled.
LAB Experimental Design & Analysis Plan and analyze Design a procedure, choose what to graph to get a straight line (linearize), then use the slope to find a quantity. Linearize: graph the quantities that make $y = mx$ a straight line.	QQT Qualitative/Quantitative Translation Explain, then verify Explain a situation in words with physics reasoning, then derive an equation that agrees with your explanation. Make sure your words and your equation predict the same trend.

◆ A time plan for the 95 minutes

MR ~25 min	TBR ~22 min	LAB ~25 min	QQT ~23 min
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A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Points are given for **derivations**. "Plugging in" without a starting equation often loses credit.

FREE-RESPONSE PLAYBOOK

05

Anatomy of a full-credit answer

Here is a short practice prompt with a full-credit answer, and the habits that earn points.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (QQT style): A battery of emf 12 V is connected to a $4\ \Omega$ resistor in series with a parallel pair of $6\ \Omega$ and $3\ \Omega$ resistors. (a) Without calculating, explain which of the two parallel resistors dissipates more power. (b) Calculate the power dissipated in the $3\ \Omega$ resistor. (c) Show that your result agrees with (a).

Full-credit response

① (a) The $3\ \Omega$ resistor. Both parallel resistors have the same potential difference, and $P = \Delta V^2/R$, so the smaller resistance dissipates more power.

② (b) Parallel: $(6 \cdot 3)/(6 + 3) = 2\ \Omega$. Total $R = 6\ \Omega$, so $I = 12/6 = 2\ \text{A}$. $\Delta V_{\text{parallel}} = 2\ \text{A} \times 2\ \Omega = 4\ \text{V}$.

③ $P_{3\Omega} = (4\ \text{V})^2/3\ \Omega \approx 5.3\ \text{W}$.

④ (c) $P_{6\Omega} = 16/6 \approx 2.7\ \text{W}$, which is less than $5.3\ \text{W}$, matching (a).

- ① **Name the shared quantity** (same ΔV in parallel) and the relationship.
- ② **Reduce the circuit** step by step and find the total current.
- ③ **Use the right equation** for the known quantities.
- ④ **Compare numerically** to tie the math to the reasoning.

Five habits for every free-response question

1

Answer every lettered part, in order, with a label.

2

Show the equation, the substitution and the units.

3

Justify with a cause $\rightarrow$ effect sentence, not a word.

4

Use the reference sheet; you don't need to memorize it.

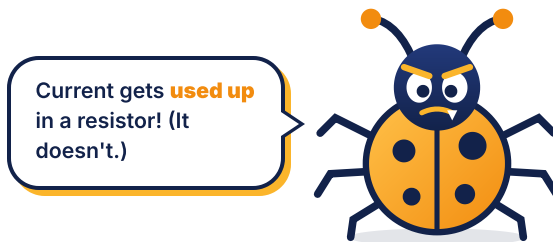
5

Blank part? Write the relationship you would use. It can earn credit.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.



01 Current "used up"

Current is the same everywhere in a series path. Resistors use up potential energy, not charge.

02 Field vs potential

Field is a vector (direction matters); potential is a scalar (just add with signs). Field can be zero where potential isn't.

03 Sign of work on a gas

In $\Delta U = Q + W$, W is work done **on** the gas. Expansion means W is negative.

04 Right-hand rule for electrons

For a negative charge, the force is opposite to what the right-hand rule gives for a positive charge.

05 Lenz's law direction

The induced current opposes the **change** in flux, not the flux itself.

06 Lens sign conventions

Real images have positive image distance; a negative magnification means the image is inverted.

07 Light bending at a boundary

Entering a higher index, light bends toward the normal and slows; its frequency stays the same.

08 Photoelectric intensity

Brighter light releases more electrons but not faster ones. Only frequency changes their maximum kinetic energy.

09 Unit conversions

Convert nm, cm, μC and eV to SI units before using equations.

10 Words and math disagree

In QQT questions, an explanation that contradicts your equation loses points on both.

Quick self-check: two identical bulbs are in series with a battery. A third identical bulb is added in parallel with one of them. Does the other bulb get brighter or dimmer? (Brighter: total resistance drops, so more current flows through it.)

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, and keep doing a few problems from older units every week.



Sep – Oct

Heat + gases



Unit 9

Ideal gases, the first law and PV diagrams, heat engines and entropy. Practice sign conventions now.

Nov – Dec

Electrostatics



Unit 10

Charges, fields, potential and capacitors. Draw field lines and equipotentials until they make sense.

Jan – Feb

Circuits +
magnetism



Units 11–12

electricity

Circuits, Kirchhoff's rules, magnetic forces and induction. One free-response question per week.

Mar – Apr

Light + modern



Units 13–15

Optics, waves, interference, photons and nuclear physics. Begin full timed sections in April.

Early May

Sharpen



Free-response drills

All four question types under timing. Review sign conventions and the traps list.

Exam day

May



AP Physics 2 exam

Sleep well, pack your calculator, and use the checklist at the end of this guide.

• The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: 21 multiple-choice questions plus 2 free-response questions	List your two weakest units
3 weeks out	Fix weak units; one of each free-response type	80%+ on unit drills
2 weeks out	Full 3-hour practice exam	Finish both sections on time
Final week	Light review: equation sheet, diagram rules, traps list	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Use $g = 9.8 \text{ m/s}^2$ unless stated. Give yourself **22 minutes** for all ten, then check page 10.

1 UNIT 9 • FIRST LAW

A gas expands at a constant pressure of $2.0 \times 10^5 \text{ Pa}$ from 0.010 m^3 to 0.030 m^3 while absorbing $10,000 \text{ J}$ of heat. What is the change in its internal energy?

- A +6,000 J
- B +4,000 J
- C +10,000 J
- D +14,000 J

2 UNIT 9 • HEAT ENGINES

A heat engine operates between reservoirs at 500 K and 300 K . What is the maximum possible efficiency?

- A 20%
- B 40%
- C 60%
- D 67%

3 UNIT 10 • ELECTRIC POTENTIAL

Charges of $+4 \text{ } \mu\text{C}$ and $-2 \text{ } \mu\text{C}$ are 0.20 m apart. What is the electric potential at the midpoint? ($k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$)

- A 0 V
- B $1.8 \times 10^5 \text{ V}$
- C $5.4 \times 10^5 \text{ V}$
- D $5.4 \times 10^6 \text{ V}$

4 UNIT 11 • CIRCUITS

A 12 V battery is connected to a $4 \text{ } \Omega$ resistor in series with a parallel pair of $6 \text{ } \Omega$ and $3 \text{ } \Omega$ resistors. What is the current through the $3 \text{ } \Omega$ resistor?

- A 0.67 A
- B 1.33 A
- C 2.0 A
- D 4.0 A

5 UNIT 12 • INDUCTION

A metal rod 0.50 m long slides at 3.0 m/s along rails, perpendicular to a 0.40 T magnetic field. What emf is induced across the rod?

- A 0.067 V
- B 0.24 V
- C 0.60 V
- D 3.75 V

6 UNIT 12 • MAGNETIC FORCE

An electron moves east through a magnetic field that points north. What is the direction of the magnetic force on the electron?

- A Up (out of the ground)
- B West
- C There is no force
- D Down (into the ground)

7 UNIT 13 • THIN LENSES

An object is 30 cm from a converging lens with a focal length of 10 cm . Which describes the image?

- A Real, inverted, 15 cm from the lens, half the size
- B Virtual, upright, 15 cm from the lens
- C Real, inverted, 30 cm from the lens, same size
- D Real, upright, 7.5 cm from the lens

8 UNIT 13 • TOTAL INTERNAL REFLECTION

Light travels from glass ($n = 1.5$) toward air. Above about what angle of incidence does total internal reflection occur?

- A 30°
- B 42°
- C 48°
- D 56°

9 UNIT 15 • PHOTOELECTRIC EFFECT

Light of wavelength 500 nm strikes a metal with a work function of 2.0 eV . What is the maximum kinetic energy of the emitted electrons? ($hc \approx 1,240 \text{ eV}\cdot\text{nm}$)

- A 2.0 eV
- B 2.48 eV
- C No electrons are emitted
- D 0.48 eV

10 UNIT 14 • INTERFERENCE

In a double-slit experiment with 600 nm light, slits 0.30 mm apart and a screen 2.0 m away, what is the spacing between bright fringes?

- A 0.4 mm
- B 2.0 mm
- C 4.0 mm
- D 40 mm

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	A • +6,000 J	Work done on the gas $W = -P\Delta V = -(2 \times 10^5)(0.020) = -4,000$ J. $\Delta U = Q + W = 10,000 - 4,000 = 6,000$ J.
2	B • 40%	Carnot efficiency $= 1 - T_C/T_H = 1 - 300/500 = 0.40$. Real engines are less efficient.
3	B • 1.8×10^5 V	Each charge is 0.10 m away. $V = k(4 \mu\text{C} - 2 \mu\text{C})/0.10 = 8.99 \times 10^9 \times 2 \times 10^{-6}/0.10 \approx 1.8 \times 10^5$ V. Potentials add as scalars with sign.
4	B • 1.33 A	Parallel pair $= 2 \Omega$; total $= 6 \Omega$; $I = 2$ A; ΔV across the pair $= 4$ V; $I_{3\Omega} = 4/3 \approx 1.33$ A.
5	C • 0.60 V	Motional emf $= BLv = 0.40 \times 0.50 \times 3.0 = 0.60$ V.
6	D • Down	For a positive charge moving east in a northward field, $F = qv \times B$ points up (east $\times$ north = up). The electron is negative, so the force is reversed: down.
7	A	$1/d_i = 1/10 - 1/30$, so $d_i = 15$ cm (real). $m = -15/30 = -0.5$: inverted and half size.
8	B • 42°	$\sin \theta_c = n_{\text{air}}/n_{\text{glass}} = 1/1.5$, so $\theta_c \approx 41.8^\circ$.
9	D • 0.48 eV	Photon energy $= 1,240/500 = 2.48$ eV. $K_{\text{max}} = 2.48 - 2.0 = 0.48$ eV.
10	C • 4.0 mm	$\Delta y = \lambda L/d = (600 \times 10^{-9})(2.0)/(0.30 \times 10^{-3}) = 4.0 \times 10^{-3}$ m $= 4.0$ mm.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Start with circuits and electric fields (Units 10–11).

10 KNOW THE TARGET

How students score

In recent years, about 72–73% of students earned a 3 or higher on AP Physics 2, and about 20–22% earned a 5. Steady preparation is what moves a score up a full point.

72–73%

of students earned a 3 or higher in recent years

20–22%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved calculator** with fresh batteries
- ☐ Plan your time: about **2 min** per multiple-choice question and **23 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ A **ruler or straightedge** for graphs and diagrams, if permitted

In the free-response section: start from a fundamental principle, draw clean labeled diagrams, keep words and equations consistent, and show every step of a derivation before plugging in numbers.



ACT SAT GO • 1-ON-1 AP TUTORING

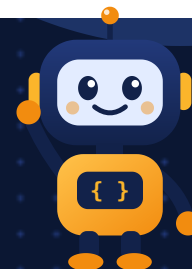
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